

lie near the secreting cell (see Figure 45.2b). In **autocrine** signaling (from the Greek *auto*, self), the secreting cells themselves are the target cells (see Figure 45.2c).

One group of local regulators are the **prostaglandins**, which are produced throughout the body and have diverse functions. In the immune system, for example, prostaglandins promote inflammation and the sensation of pain in response to injury. Drugs that block prostaglandin synthesis, such as aspirin and ibuprofen, prevent these activities, producing both anti-inflammatory and pain-relieving effects.

Prostaglandins are modified fatty acids. Many other local regulators are polypeptides, including cytokines, which enable immune cell communication (see Figure 43.16 and Figure 43.17), and growth factors, which promote cell growth, division, and development.

Some local regulators, such as **nitric oxide (NO)**, are gases. When the level of oxygen in the blood falls, endothelial cells in blood vessel walls synthesize and release NO. After diffusing into the surrounding smooth muscle cells, NO activates an enzyme that relaxes the cells. The result is vasodilation, which increases blood flow to tissues.

In human males, NO's ability to promote vasodilation enables sexual function by increasing blood flow into the penis, producing an erection. The drug Viagra (sildenafil citrate), a treatment for male erectile dysfunction, sustains an erection by prolonging activity of the NO response pathway.

Synaptic and Neuroendocrine Signaling

Secreted molecules are essential for the function of the nervous system. Neurons communicate with target cells, such as other neurons and muscle cells, via specialized junctions called synapses. At most synapses, neurons secrete molecules called **neurotransmitters** that diffuse a very short distance (a fraction of a cell diameter) and bind to receptors on the target cells (see Figure 45.2d). Such *synaptic signaling* is central to sensation, memory, cognition, and movement (as we'll explore in Chapters 48–50).

In *neuroendocrine signaling*, neurons called neurosecretory cells secrete **neurohormones**, which diffuse from nerve cell endings into the bloodstream (see Figure 45.2e). One example of a neurohormone is antidiuretic hormone, which functions in kidney function and water balance as well as courtship behavior. Many neurohormones regulate endocrine signaling, as we'll discuss later in this chapter.

▼ **Figure 45.3 Signaling by pheromones.** Using their lowered antennae, these Asian army ants (*Leptogenys distinguenda*) carry pupae and larvae along a pheromone-marked trail to a new nest site.



Signaling by Pheromones

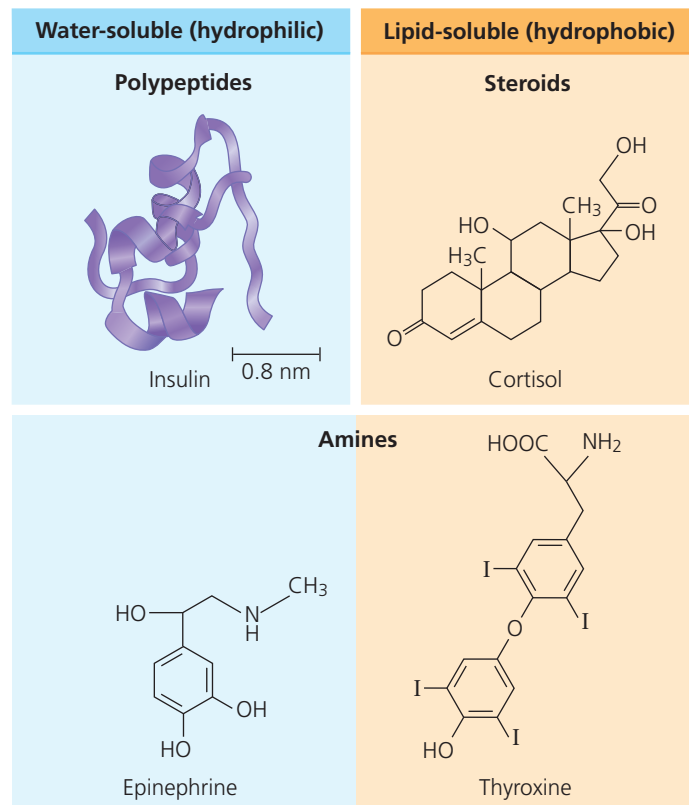
Not all secreted signaling molecules act within the body. Members of a particular animal species sometimes communicate with each other via **pheromones**, chemicals that are released into the external environment. For example, when a foraging ant discovers a new food source, it marks its path back to the nest with a pheromone. Ants also use pheromones for guidance when a colony migrates to a new location (Figure 45.3).

Pheromones serve a wide range of functions that include defining territories, warning of predators, and attracting potential mates. The polyphemus moth (*Antheraea polyphemus*) provides a noteworthy example: The sex pheromone released into the air by a female enables her to attract a male of the species from up to 4.5 km away. You'll read more about pheromone function when we take up the topic of animal behavior in Chapter 51.

Chemical Classes of Hormones

Hormones fall into three major chemical classes: polypeptides, steroids, and amines (Figure 45.4). The hormone insulin, for example, is a polypeptide that contains two chains in its active form. Steroid hormones, such as cortisol, are lipids

▼ **Figure 45.4 Variation in hormone solubility and structure.**



MAKE CONNECTIONS Cells synthesize epinephrine from the amino acid tyrosine. On the structure of epinephrine shown above, draw a circle around the portion of the molecule corresponding to the R group of tyrosine (see Figure 5.14).